

ASSESSMENT REPORT

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Assessment : JavaScript Assessment -REG - React JS - 04 May 2026
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TOTAL SCORE	QUESTIONS	TIME TAKEN	RANK
78.6%	3/5	74 min	#1/6

PERFORMANCE ANALYTICS

Coding Questions

Total : 5
Correct : 3
Incorrect : 2

DETAILED QUESTION REVIEW

Q1

Question : Predict the console output order for the following JavaScript code snippet. Your function `predictOrder1` should simulate the execution and return an array of strings representing the order of logs.

```
```javascript
console.log('A');
setTimeout(() => console.log('B'), 0);
Promise.resolve().then(() => console.log('C'));
console.log('D');
```
```

Your Answer : `const readline = require('readline');`

```

const rl = readline.createInterface({
  input: process.stdin,
  output: process.stdout
});

function predictOrder1() {
  // your code here
  // The function should return an array of strings,
  // where each string is a log message in the order it would appear.
  // For example: ['A', 'D', 'C', 'B']

  let order=[];
  order.push('A');
  order.push('D');
  order.push('C');
  order.push('B');
  return order;
}

rl.on('line', (line) => {
  const result = predictOrder1();
  console.log(result.join(','));
  rl.close();
});

```

Correct Answer : const readline = require('readline');

```

const rl = readline.createInterface({
  input: process.stdin,
  output: process.stdout
});

function predictOrder1() {
  const logs = [];
  const microtasks = [];
  const macrotasks = [];

  logs.push('A');

  macrotasks.push('B'); // setTimeout callback

  microtasks.push('C'); // Promise.then callback

  logs.push('D');

  return [...logs, ...microtasks, ...macrotasks];
}

```

```

rl.on('line', (line) => {
  const result = predictOrder1();
  console.log(result.join(','));
  rl.close();
});

```

Points : 100/100

Status : Correct

Code Logic Score : 100.0

Space Complexity : O(1)

Time Complexity : O(1)

TEST CASE RESULTS

| S.No | Input | Expected Output | Actual Output | Status |
|------|-------|-----------------|---------------|--------|
| 1 | start | A,D,C,B | A,D,C,B | PASS |
| 2 | start | A,D,C,B | A,D,C,B | PASS |
| 3 | start | A,D,C,B | A,D,C,B | PASS |
| 4 | start | A,D,C,B | A,D,C,B | PASS |
| 5 | start | A,D,C,B | A,D,C,B | PASS |
| 6 | start | A,D,C,B | A,D,C,B | PASS |
| 7 | start | A,D,C,B | A,D,C,B | PASS |
| 8 | start | A,D,C,B | A,D,C,B | PASS |
| 9 | start | A,D,C,B | A,D,C,B | PASS |
| 10 | start | A,D,C,B | A,D,C,B | PASS |
| 11 | start | A,D,C,B | A,D,C,B | PASS |
| 12 | start | A,D,C,B | A,D,C,B | PASS |
| 13 | start | A,D,C,B | A,D,C,B | PASS |
| 14 | start | A,D,C,B | A,D,C,B | PASS |
| 15 | start | A,D,C,B | A,D,C,B | PASS |

Q2

Question : Given an array of strings strs, group the anagrams together. You can return the answer in any order.

Your Answer : `const readline = require('readline');`

```

const rl = readline.createInterface({
  input: process.stdin,

```

```

    output: process.stdout
  });

let inputLines = [];
rl.on('line', (line) => {
  inputLines.push(line);
});

rl.on('close', () => {
  const strs = inputLines;
  const result = groupAnagrams(strs);
  result.forEach(group => {
    console.log(group.sort().join(' ')); // Sort each group for consistent output
  });
});

function groupAnagrams(strs) {

  const map= new Map();
  for (const word of strs){
    const key=word.split("").sort().join("");
    if (!map.has(key)){
      map.set(key,[]);
    }
    map.get(key).push(word);
  }
  return Array.from(map.values());
}

```

Correct Answer : const readline = require('readline');

```

const rl = readline.createInterface({
  input: process.stdin,
  output: process.stdout
});

let inputLines = [];
rl.on('line', (line) => {
  inputLines.push(line);
});

rl.on('close', () => {
  const strs = inputLines;
  const result = groupAnagrams(strs);
  result.forEach(group => {
    console.log(group.sort().join(' ')); // Sort each group for consistent output
  });
});

```

```

    });
  });

function groupAnagrams(strs) {
  const map = new Map();

  for (const str of strs) {
    const sortedStr = str.split('').sort().join('');
    if (map.has(sortedStr)) {
      map.get(sortedStr).push(str);
    } else {
      map.set(sortedStr, [str]);
    }
  }

  return Array.from(map.values());
}

```

Points : 0/100

Status : **Incorrect**

TEST CASE RESULTS

Compilation error, No test cases were passed

Q3

Question : Given an array `nums` of size `n`, return the majority element. The majority element is the element that appears more than $n / 2$ times. You may assume that the majority element always exists in the array.

Your Answer : `const readline = require('readline');`

```

const rl = readline.createInterface({
  input: process.stdin,
  output: process.stdout
});

```

```

function majorityElement(nums) {
  // your code here
  let candidate=null;
  let count=0;
  for(let num of nums){
    if(count===0){
      candidate=num;
    }
    if(num===candidate){
      count++;
    }
  }
}

```

```

}else{
count--;
}
}
return candidate;
}

```

```

let inputLines = [];
rl.on('line', (line) => {
  inputLines.push(line);
});

```

```

rl.on('close', () => {
  const nums = inputLines[0].split(' ').map(Number);
  const result = majorityElement(nums);
  console.log(result);
});

```

Correct Answer : const readline = require('readline');

```

const rl = readline.createInterface({
  input: process.stdin,
  output: process.stdout
});

```

```

function majorityElement(nums) {
  let candidate = 0;
  let count = 0;

  for (let i = 0; i < nums.length; i++) {
    if (count === 0) {
      candidate = nums[i];
      count = 1;
    } else if (nums[i] === candidate) {
      count++;
    } else {
      count--;
    }
  }
  return candidate;
}

```

```

let inputLines = [];
rl.on('line', (line) => {
  inputLines.push(line);
});

```

```

rl.on('close', () => {
  const nums = inputLines[0].split(' ').map(Number);
  const result = majorityElement(nums);
  console.log(result);
});

```

Points : 100/100

Status : Correct

Code Logic Score : 95.0

Space Complexity : O(1)

Time Complexity : O(n)

TEST CASE RESULTS

| S.No | Input | Expected Output | Actual Output | Status |
|------|--|-----------------|---------------|--------|
| 1 | 3 2 3 | 3 | 3 | PASS |
| 2 | 2 2 1 1 1 2 2 | 2 | 2 | PASS |
| 3 | 1 | 1 | 1 | PASS |
| 4 | 7 7 7 7 7 7 7 | 7 | 7 | PASS |
| 5 | 1 2 1 2 1 | 1 | 1 | PASS |
| 6 | 6 5 5 | 5 | 5 | PASS |
| 7 | 10 9 10 9 10 | 10 | 10 | PASS |
| 8 | 1 1 2 2 1 1 2 2 1 | 1 | 1 | PASS |
| 9 | 5 5 5 1 1 1 5 | 5 | 5 | PASS |
| 10 | 3 3 3 3 2 2 2 | 3 | 3 | PASS |
| 11 | 1 2 3 4 5 5 5 5 5 | 5 | 5 | PASS |
| 12 | 1 1 1 1 1 2 3 4 5 | 1 | 1 | PASS |
| 13 | 10 20 10 20 10 20 10 | 10 | 10 | PASS |
| 14 | 1 1 1 1 1 1 1 1 1 2
3 4 5 6 7 8 9 10 | 1 | 1 | PASS |
| 15 | 100 100 100 100 100
100 100 100 100 100
1 2 3 4 5 6 7 8 9 10 | 100 | 100 | PASS |

Q4

Question : Write a function `mergeAndSumUnique` that accepts an arbitrary number of arrays of numbers using the rest parameter syntax. The function should merge all arrays, remove duplicate numbers, and then return the sum of the unique numbers. Use ES6 array methods like `Set` and `reduce`.

```

Your Answer : const mergeAndSumUnique = (...arrays) => {
  // your code here
  return [...new Set(arrays.flat())].reduce((sum,num)=> sum + num,0);
};

const readline = require('readline');
const rl = readline.createInterface({
  input: process.stdin,
  output: process.stdout
});

let inputLines = [];
rl.on('line', (line) => {
  inputLines.push(line);
});

rl.on('close', () => {
  const arrays = inputLines.map(line => line.split(' ').map(Number));
  console.log(mergeAndSumUnique(...arrays));
});
Correct Answer : const mergeAndSumUnique = (...arrays) => {
  const allNumbers = [].concat(...arrays);
  const uniqueNumbers = new Set(allNumbers);
  return Array.from(uniqueNumbers).reduce((sum, num) => sum + num, 0);
};

const readline = require('readline');
const rl = readline.createInterface({
  input: process.stdin,
  output: process.stdout
});

let inputLines = [];
rl.on('line', (line) => {
  inputLines.push(line);
});

rl.on('close', () => {
  const arrays = inputLines.map(line => line.split(' ').map(Number));
  console.log(mergeAndSumUnique(...arrays));
});
Points : 100/100
Status : Correct
Code Logic Score : 100.0
Space Complexity : O(N)

```


Time Complexity : $O(N)$

TEST CASE RESULTS

| S.No | Input | Expected Output | Actual Output | Status |
|------|----------------------------------|-----------------|---------------|--------|
| 1 | 1 2 3
3 4 5
5 6 7 | 28 | 28 | PASS |
| 2 | 10 20
20 30
30 40 | 100 | 100 | PASS |
| 3 | 1 1 1
2 2 2 | 3 | 3 | PASS |
| 4 | 5
5
5 | 5 | 5 | PASS |
| 5 | 1 2 3 4 5 | 15 | 15 | PASS |
| 6 | 0
0
0 | 0 | 0 | PASS |
| 7 | 100 200
300 400 | 1000 | 1000 | PASS |
| 8 | 7 8 9
9 10 11 | 45 | 45 | PASS |
| 9 | 1 2
3 4
5 6
7 8
9 10 | 55 | 55 | PASS |
| 10 | 1 3 5
2 4 6 | 21 | 21 | PASS |
| 11 | 10 10 10
20 20 20
30 30 30 | 60 | 60 | PASS |
| 12 | 1 2 3 4 5 6 7 8 9 10 | 55 | 55 | PASS |
| 13 | 1 1 2 2 3 3 | 6 | 6 | PASS |
| 14 | 10 20 30
10 20 30 | 60 | 60 | PASS |
| 15 | 1
2
3
4
5 | 15 | 15 | PASS |

Q5

Question : Given an array of strings `strs`, group the anagrams together. You can return the answer in any order. An Anagram is a word or phrase formed by rearranging the letters of a different word or phrase, typically using all the original letters exactly once.

Your Answer : `const readline = require('readline');`

```
const rl = readline.createInterface({
  input: process.stdin,
  output: process.stdout
});

let inputLines = [];
rl.on('line', (line) => {
  inputLines.push(line);
});

rl.on('close', () => {
  const n = parseInt(inputLines[0]);
  const strs = [];
  for (let i = 1; i <= n; i++) {
    strs.push(inputLines[i]);
  }

  const result = groupAnagrams(strs);
  // Sort groups and elements within groups for consistent output
  result.forEach(group => group.sort());
  result.sort((a, b) => a[0].localeCompare(b[0]));

  result.forEach(group => {
    console.log(group.join(' '));
  });
});

function groupAnagrams(strs) {
  // your code here
  const map = new Map();
  for (const str of strs) {
    const sorted = str.split('').sort().join('');
    if (!map.has(sorted)) {
      map.set(sorted, []);
    }
    map.get(sorted).push(str);
  }
  return Array.from(map.values());
}
```

Correct Answer : const readline = require('readline');

```
const rl = readline.createInterface({
  input: process.stdin,
  output: process.stdout
});

let inputLines = [];
rl.on('line', (line) => {
  inputLines.push(line);
});

rl.on('close', () => {
  const n = parseInt(inputLines[0]);
  const strs = [];
  for (let i = 1; i <= n; i++) {
    strs.push(inputLines[i]);
  }

  const result = groupAnagrams(strs);
  // Sort groups and elements within groups for consistent output
  result.forEach(group => group.sort());
  result.sort((a, b) => a[0].localeCompare(b[0]));

  result.forEach(group => {
    console.log(group.join(' '));
  });
});

function groupAnagrams(strs) {
  const anagramMap = new Map();

  for (const str of strs) {
    const sortedStr = str.split('').sort().join('');
    if (anagramMap.has(sortedStr)) {
      anagramMap.get(sortedStr).push(str);
    } else {
      anagramMap.set(sortedStr, [str]);
    }
  }

  return Array.from(anagramMap.values());
}
```

Points : 93/100

Status : **Incorrect**

Code Logic Score : 100.0
Space Complexity : $O(N * L)$
Time Complexity : $O(N * L \log L)$

TEST CASE RESULTS

| S.No | Input | Expected Output | Actual Output | Status |
|------|---|---------------------------------|---------------------------------|--------|
| 1 | 4
eat
tea
tan
nat | eat tea
nat tan | eat tea
nat tan | PASS |
| 2 | 3
hello
world
apple | apple
hello
world | apple
hello
world | PASS |
| 3 | 1
a | a | a | PASS |
| 4 | 0 | | | PASS |
| 5 | 2
a
b | a
b | a
b | PASS |
| 6 | 5
listen
silent
enlist
cat
act | act cat
enlist listen silent | act cat
enlist listen silent | PASS |
| 7 | 3
rat
tar
art | art rat tar | art rat tar | PASS |
| 8 | 4
flower
flow
wolf
fowl | flow
flower
fowl wolf | flow fowl wolf
flower | FAIL |
| 9 | 2
abc
bca | abc bca | abc bca | PASS |
| 10 | 2
xyz
yxz | xyz yxz | xyz yxz | PASS |

| | | | | |
|----|---|----------------------------------|----------------------------------|------|
| 11 | 5
stop
pots
tops
spot
post | post pots spot stop
tops | post pots spot stop
tops | PASS |
| 12 | 3
aa
a
b | a
aa
b | a
aa
b | PASS |
| 13 | 4
aba
aab
bba
bab | aab aba
bab bba | aab aba
bab bba | PASS |
| 14 | 2
apple
banana | apple
banana | apple
banana | PASS |
| 15 | 6
code
doce
ecod
aced
dace
caed | aced caed dace
code doce ecod | aced caed dace
code doce ecod | PASS |